Lu Sun, and many more.

A Notebook on Algebra

To my family, friends and communities members who have been dedicating to the presentation of this notebook, and to all students, researchers and faculty members who might find this notebook helpful.

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Foreword

I was not a big supporter for sharing notebooks. Notebooks, in my view, had been private and practically useless to anyone other than their owner.

It traces back to the summer of 2009 when I started my third year as a high school student. In the end of August when the results of Gaokao (National College Entrance Examination of China, annually held in July) came out, photocopy shops would start selling notebooks photocopies that they claimed to be from the top scorers of the exam. Much curious as I was about what these notebooks look like, never have I purchased any of them, and nor expected myself to actually learn anything from them. In fact, many of these notebooks were more difficult to read than the textbooks. I guess we could not blame the top scorers for being so smart that they sometimes made things extremely brief or overwhelmingly comprehensive. Not to mention, why would I want to adapt to notebooks of others when I had my own notebooks which in my opinion should be just as good as theirs.

However, my mind changed after becoming an undergraduate student in 2010 with the school of electrical engineering. There were so many engineering modules, and as an unfortunate result it was very challenging to dig deep into all of them evenly. I naturally put less effort on "side" modules such as material science than my major's core modules such as analog electronics. The situation became worse when I started pursuing my Ph.D. in 2014. As I had to focus on specific research areas entirely, I could hardly split enough time to systematically study those less-relevant yet interesting contents. For example, my research area was on robust statistics, state estimation and adaptive control, and I could not spend as much time as I wished on graph theory. I needed a way to quickly scratch the surface of some areas, and became decently familiar with them. There was no intention to become an expert on these areas by all means. I just wanted to understand what they are, and see whether they could have inspired my own research.

To make a difference, I forced myself to read articles beyond my comfort zone and it ended up motivating me to take notes to consolidate the knowledge. I started with easy subjects. My very first notebook was on Numerical Analysis, an entrance-level module for engineering background graduate students. Revisiting that module did not bring much benefit to my research, but it served as a good starting point. Following that were some more math notebooks, then a few on computing, operating systems and databases, and then on data science, and then on ontology.

Eventually at one point, it suddenly came to me: these notebooks that I

had taken while learning these subjects as an armature, they were exactly the materials I needed, quoting myself, to "quickly scratch the surface of some areas, and became decently familiar with them". Well, in that case, why not digitizing them, making them accessible online and open-source and letting everyone read and edit it?

Hence, this is where we are.

Similar with most open-source software, these notebooks do not come with any "warranty" of any kind, meaning that there is no guarantee that everything in the notebooks is correct. Always bear in mind that the notebooks are not peer reviewed. **Do NOT cite this notebook or any notebooks in this series in your academic research paper!** If you find anything helpful here and you want to use it in your research, please trace back to the origin of the knowledge and confirm by yourself.

This notebook is suitable as:

- a quick reference guide;
- a brief introduction for beginners of an area;
- a "cheat sheet" for students or teachers for an entrance-level course.

This notebook is NOT suitable as:

- a direct research reference;
- a replacement of the corresponding textbook.

Again, this notebook is NOT peer reviewed. It is meant to be easy to read, not to be comprehensive and very rigorous.

Although this notebook is open-source, the reference materials of this notebook, including textbooks, journal papers, conference proceedings, etc., may not be open-source. Very likely many of these reference materials are licensed or copyrighted. Please legitimately access these materials and properly use them, should you decided to trace the origin of the knowledge.

Some of the figures in this notebook are plotted using Excalidraw, a very convenient tool to emulate hand drawings. The Excalidraw project can be found on GitHub, *excalidraw/excalidraw*. Other figures may come from MATLAB, R, Python, and other computation engines. The source code to reproduce the results are likely to be included in the same repository of the notebook, but there might be exceptions.

This work might have benefited from the assistance of large language models, which are used exclusively for editing purposes such as correcting grammar and rephrasing sentences, without introducing new content, generating innovative information, or changing the original intent of the text.

Preface

This notebook is on algebra.

The first part of the notebook is about linear algebra, one of the first few modules a science and engineering undergraduate student would take in their first semester in the collage. It is absolutely the fundamental of almost all the mathematical tools he would use in the future.

The second part of the notebook is about abstract algebra, a far more advanced topic and yet still a lot of fun to learn. The third part of the notebook is on number theory, which is a natural expansion to abstract algebra.

The key reference of this notebook is listed below. During the development of the notebook, this list may become longer and longer.

A list of key references of this notebook is given below. These materials are frequently cited in the notebook. For convenience, they are not given in the Reference section but listed here instead, whereas otherwise I will have to cite them everywhere in the text.

- Lay, David C., Steven R. Lay, and Judi J. McDonald. "Linear Algebra and its Applications", 5/E, Pearson, 2016
- Jacobson, Nathan. Basic algebra (I and II). Courier Corporation, 2012.

Other reference materials include a series of videos by *Meditation Math* on Bilibili regarding linear algebra, which I have found not just entertaining but also very inspiring.

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Part I

Linear Algebra

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Linear Algebra in a Nutshell

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This chapter gives an overview of linear algebra, including its scope and how the different terminologies and tools used in linear algebra are related to each other. The purpose of this chapter is to help the reader build the bird's-eye view and make them feel more under control when encountering the abstract concepts in later chapters.

Consider this chapter as additional help. Everything in this chapter will be formally introduced in later chapters at some point. The reader is free to skip this chapter should such help not required.

1.1 Algebra

The term "algebra" has a rich variety of meanings. In the context of elementary mathematics, **algebra** may be viewed as a way of studying quantities and their relationships using symbols rather than specific numerical values. By introducing symbols, the focus shifts from solving a particular problem to discovering general principles that apply to many problems at once.

The following example illustrates this idea.

A Motivating Example of Algebra

Consider the problem of finding the area of an annulus whose outer radius is 2 and whose inner radius is 1.

The area can be computed directly by subtracting the area of the smaller circle from that of the larger circle:

$$\begin{aligned} S &= 2^2\pi - 1^2\pi \\ &= 3\pi. \end{aligned}$$

Surprisingly, when I first encountered this problem as a primary school student, I arrived at a different approach. Imagine a line segment of length $2-1=1$ rotating about the centre of the annulus. As it rotates, it sweeps out the entire annular region, as illustrated in Fig. 1.1. The bold black segment represents the rotating line segment, while the shaded red region represents the area that has already been swept out.

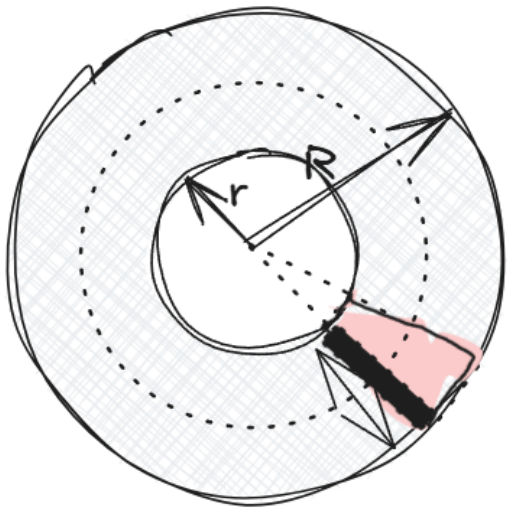


FIGURE 1.1: Calculating the area of an annulus by sweeping a line segment.

The total area swept out by the segment should therefore be equal to the length of the segment multiplied by the distance it travels. A similar idea is often used when explaining the area of a parallelogram, where a line segment sweeps out a region through horizontal or vertical translation.

The difficulty in the present problem is that the segment rotates rather than translates. During the rotation, its two endpoints travel along circles of different circumferences. It is therefore not immediately clear what distance should be used in the calculation. My intuition was that the appropriate distance should be the one traveled by the midpoint of the segment, which traces out the dashed circle shown in the figure. At the time, I had no proof that this choice was correct; it simply seemed the most natural one.

Using this idea, the area of the annulus is

$$\begin{aligned} S &= (2-1) \times (\pi(2+1)) \\ &= 3\pi, \end{aligned}$$

where $(2-1)$ is the length of the segment and $\pi(2+1)$ is the distance traveled by its midpoint given by the dashed circle in Figure 1.1. Since $(2+1)$ is the diameter of the dashed circle, $\pi(2+1)$ is precisely its circumference. The result agrees with the direct calculation given earlier.

We want to know whether my method would always produce the correct answer for any annulus with a different pair of inner and outer radius. Looking back at the solution and with the more abstract and powerful tool of algebra, it can be done easily. Let R and r denote the outer and inner radius of the annulus respectively. Using the first approach, we obtain

$$S = \pi R^2 - \pi r^2.$$

Using the second approach, the length of the rotating segment is $R-r$, while the distance traveled by its midpoint is the circumference of a circle with diameter $R+r$. Hence,

$$\begin{aligned} S &= (R-r)\pi(R+r) \\ &= \pi(R^2 - r^2) \\ &= \pi R^2 - \pi r^2. \end{aligned}$$

The two approaches therefore give exactly the same result.

More importantly, by replacing the specific numbers 2 and 1 with the symbols R and r, we have transformed a particular arithmetic problem into a general statement. Instead of obtaining a result for only one annulus, we have derived a formula that applies to every annulus. The symbols allow us to identify the underlying relationship

$$R^2 - r^2 = (R-r)(R+r),$$

which explains why the two seemingly different methods lead to the same answer.

This illustrates one of the central ideas of algebra. Rather than studying individual numerical examples, we use symbols to uncover general patterns and relationships that hold in a wide range of situations. By using the symbols, the problem becomes abstract to some extent, and in return we have a more generalized solution and more insightful understanding about the problem.

1.2 Linear Algebra

Linear algebra is a subset of algebra. It studies a particular portion of algebra that concerns with vector and linear transformation, both of which introduced in the next sections.

1.2.1 Vector and Vector Space

A vector is an element of a vector space. A vector space is an algebraic structure consisting of a non-empty set of objects, called vectors, together with two operations, vector addition and scalar multiplication. These operations satisfy a collection of axioms, commonly known as the vector space axioms. The abstract definition of a vector space belongs to the subject of abstract algebra and will be discussed in greater detail in Part II of this book.

For the moment, it is sufficient to think of a vector as an ordered list of real numbers of fixed length. Two operations can be performed on such vectors.

Vector addition. The sum of two vectors is obtained by adding their corresponding components. Let $x = [x_1, \ldots, x_n]^T$, $y = [y_1, \ldots, y_n]^T$. Their sum is given by

$$x + y \quad = \quad \begin{bmatrix} x_1 + y_1 \\ \vdots \\ x_n + y_n \end{bmatrix}.$$

Vector subtraction is defined similarly by subtracting corresponding components.

Scalar multiplication. Multiplying a vector by a scalar is done by multiplying each component by that scalar. Let $x = [x_1, \ldots, x_n]^T$ and let c be a scalar. Then

$$cx \quad = \quad \begin{bmatrix} cx_1 \\ \vdots \\ cx_n \end{bmatrix}.$$

At this stage, one may think of a vector space as a collection of vectors of the same dimension together with the operations of vector addition and scalar multiplication. This description captures the examples that will be studied in the first part of this book, although the formal definition introduced later is considerably more general.

Readers with a computer science or engineering background may find it helpful to think of vectors as a generalization of scalar numbers. A scalar may be regarded as a vector of dimension 1. From this perspective, vector addition and scalar multiplication naturally extend the familiar arithmetic operations on real numbers.

Notice that a vector space, like any algebraic structure, must be closed to the operations defined on them. In the case of vector space, this would mean that if V is a vector space, $\forall x, y \in V$, $x + y \in V$, and $\forall x \in V$, $cx \in V$. Examples of vector spaces include $\mathbb{R}^n$ and $\{[x_1, x_2]^T | x_1 = 2x_2\}$, etc. It can be easily proved that the above examples are all close with respect to vector addition and scalar multiplication. On the other hand, $\mathbb{Z}^n$ is not a vector space because it may be closed to scalar multiplications with c a fraction. Similarly, $\mathbb{R}^n_+$ is not a vector space either, as it is not closed to scalar multiplications with negative c.

1.2.2 Linear Transformation

A **transformation** is a mapping that takes an input and produces an output according to a specified rule. A **linear transformation** is a transformation $T : V \to W$ between two vector spaces that preserves vector addition and scalar multiplication which is known as the **superposition principle of linear system**. That means for all $v, v_1, v_2 \in V$ and scalar c,

$$\begin{aligned} T(v_1 + v_2) &= T(v_1) + T(v_2), \\ T(cv) &= cT(v). \end{aligned}$$

Superposition principle of linear system implies that a linear transformation preserves the linear structure of a vector space.

An example of linear transformation is given below.

Example of Linear Transformation

Let x_1, x_2 denote the number of chicken and rabbit in a cage respectively, and y_1, y_2 the total number of heads and feet respectively. Note that a chicken has 1 head and 2 feet whereas a rabbit, 1 head and 4 feet. Clearly,

$$y_1 = x_1 + x_2, \tag{1.1a}$$
$$y_2 = 2x_1 + 4x_2. \tag{1.1b}$$

Let $x = [x_1, x_2]^T$ and $y = [y_1, y_2]^T$, and denote the transformation $y = f(x)$. It can be proved that this is a linear transformation.

The proof of the system described by (1.1) being a linear transformation is obvious and hence not given here. As will be introduced later in Section 1.3, this system can be denoted by

$$\begin{bmatrix} 1 & 1 \\ 2 & 4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} y_1 \\ y_2 \end{bmatrix},$$

or equivalently

$$\begin{aligned}\left(\begin{bmatrix} x_1 & x_2 \end{bmatrix}\begin{bmatrix} 1 & 2 \\ 1 & 4 \end{bmatrix}\right)^T &= \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}, \\ x_1\begin{bmatrix} 1 \\ 2 \end{bmatrix} + x_2\begin{bmatrix} 1 \\ 4 \end{bmatrix} &= \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}.\end{aligned}$$

1.2.3 Linear Algebra

In a nutshell, **linear algebra** is the branch of mathematics that studies vectors, vector spaces and linear transformations. The goal of studying linear algebra is to understand the properties of the vector spaces and linear transformations of different kinds, and how they can be used to solve varieties of problems in science and engineering.

For example, given system (1.1) and a particular set of $y = [y_1, y_2]^T$, and we want to find such x that produces y. This clearly falls into the scope of linear algebra. In the primary school, this is also known as "solving a system of linear equations in n unknowns" where $n = 2$. In linear algebra, we would like to not only solve x but also understand deeper about the relation of x and y and the conditions whether there would be a solution.

And of course, solving linear equation is nothing but one of the many use cases of linear algebra, some of which far more complicated and abstract.

1.3 Matrix

There are at least two ways of interpreting matrix. A matrix can be seen as a representation of a linear transformation. A matrix can also be seen as a collection of (base) vectors of a vector space. In this section, the two interpretations and their relations are discussed.

1.3.1 Representation of Linear Transformation

One may think of matrix as a notation of a linear transformation on a vector space. Each deterministic linear transformation of such kind is corresponding with a matrix.

1.3.2 Base Vectors of a Vector space

TBA.

2

Vector, Matrix and Linear System

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This chapter studies the concepts and basic properties of linear equations and linear systems. They serve as the basis of linear algebra.

2.1 Linear System

This section introduces linear equation and linear system, their scalar and vector notations, and their solution set.

2.1.1 Linear Equation

A **linear equation** of variables x_i refers to the equation of the following form

$$a_1x_1 + \ldots + a_nx_n = b$$

where i is the index $i = 1, \ldots, n$ with $n \geq 1$ being the total number of variables of x_i. Parameters a_i and b are real or complex constant values, where a_i are known as the **coefficients** of the equation.

A system of linear equations on the same variable set is known as a **linear system**. For example,

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \ldots + a_{1n}x_n &= b_1 \\ &\vdots \\ a_{m1}x_1 + a_{m2}x_2 + \ldots + a_{mn}x_n &= b_m \end{aligned}$$

is a linear system of variables x_i.

A linear system has solution(s) if there is at least one set of variables x_i that fulfills all the linear equations in the system. However, this is not always the case. In practice, a linear system may

- have no solution;
- have one unique solution;
- have infinite number of solutions.

The solution(s) to a linear system forms its **solution set**. The cardinality of the solution set of a linear system can be 0, 1 or infinity.

Can a linear system have more than two finite number of solutions?

A linear system cannot have more than two finite number of solutions. This can be proved by contradiction.

Assume that a linear system has a finite number of $N \geq 2$ solutions. From those solutions, select two distinct set of solutions x_i^a and x_i^b. Since the solutions are distinct, at least one of them is non-zero. Without loosing generality, let us assume that x_i^a is an non-zero solution.

Substituting x_i^a into the linear system gives

$$\begin{aligned} a_{11}x_1^a + a_{12}x_2^a + \ldots + a_{1n}x_n^a &= b_1 \\ &\vdots \\ a_{m1}x_1^a + a_{m2}x_2^a + \ldots + a_{mn}x_n^a &= b_m \end{aligned}$$

Multiplying p on both side of the equations gives

$$\begin{aligned} a_{11}px_1^a + a_{12}px_2^a + \ldots + a_{1n}px_n^a &= pb_1 \\ &\vdots \\ a_{m1}px_1^a + a_{m2}px_2^a + \ldots + a_{mn}px_n^a &= pb_m \end{aligned}$$

Substituting x_i^b into the linear system and multiplying $(1-p)$ on both side gives

$$\begin{aligned} a_{11}(1-p)x_1^b + a_{12}(1-p)x_2^b + \ldots + a_{1n}(1-p)x_n^b &= (1-p)b_1 \\ &\vdots \\ a_{m1}(1-p)x_1^b + a_{m2}(1-p)x_2^b + \ldots + a_{mn}(1-p)x_n^b &= (1-p)b_m \end{aligned}$$

Adding the two set of equations gives

$$\begin{aligned} a_{11}\left[px_1^a + (1-p)x_1^b\right] + \ldots + a_{1n}\left[px_n^a + (1-p)x_n^b\right] &= b_1 \\ &\vdots \\ a_{m1}\left[px_1^a + (1-p)x_1^b\right] + \ldots + a_{mn}\left[px_1^a + (1-p)x_1^b\right] &= b_m \end{aligned}$$

Therefore, $px_i^a + (1-p)x_i^b$ must also be a solution to the original linear system.

Let $p \in P = \{0, \frac{1}{N}, \frac{2}{N} \ldots, \frac{N}{N}\}$. The cardinality of P is $N+1$. It can be easily proved that with non-zero x_i^a, $px_i^a + (1-p)x_i^b$ are distinct values for each different values p taken from P. All of these $N+1$ distinct $px_i^a + (1-p)x_i^b$ with different p are solutions to the original linear system. This contradicts with the assumption that the linear system has a finite number of N solutions.

A linear system that has solution(s) is said to be **consistent**, whereas a linear system with no solution is said to be **inconsistent**. Two linear systems are said to be **equivalent** if they have the same solution set.

2.1.2 Matrix Notation

A linear system

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \ldots + a_{1n}x_n &= b_1 \\ &\vdots \\ a_{m1}x_1 + a_{m2}x_2 + \ldots + a_{mn}x_n &= b_m \end{aligned}$$

can be denoted by

$$Ax = b \tag{2.1}$$

where x is a $n \times 1$ vector, A a $m \times n$ **coefficient matrix**, and b a $m \times 1$ vector given by

$$\begin{aligned} x &= \begin{bmatrix} x_1 & \ldots & x_n \end{bmatrix}^T \\ A &= \begin{bmatrix} a_{11} & \ldots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{m1} & \ldots & a_{mn} \end{bmatrix} \\ b &= \begin{bmatrix} b_1 & \ldots & b_m \end{bmatrix}^T \end{aligned}$$

2.1.3 Solution to a Linear System

Matrix A and vector b contains all the information necessary to solve the linear system. **Elimination algorithm** can be used to find the solution set of a linear system. The details are too basic hence not giving in this notebook.

As mentioned earlier, a linear system may have zero, one or infinite solutions. It is interesting to see how the numbers of solutions are determined by (certain patterns) in A and b. Detailed discussions are given in later part of this notebook. A quick review is given below.

There are mainly the following factors that decide the number of solutions of a linear system.

- For the system to have solution(s), the linearly dependent equations, if any, must be consistent;
- For a system that has solution(s),
  - For a system to have one unique solution, the number of linearly independent equations must match the number of variables.
  - For a system to have infinite solutions, the number of linearly independent equations must be smaller than the number of variables n.

Notice that it is impossible to have more linearly independent equations than the number of the variables.

Consider the **homogeneous system** of (2.1) which is given by

$$Ax = 0$$

with $b = 0$ substituted into the original linear system. This can be taken as a special case of (2.1). It is obvious that the system must be consistent, and at least the zero vector $x = 0$ is a solution. The question is whether $x = 0$ is the unique solution. This is determined by the number of linearly independent rows of A, and whether it matches with the number of the independent variables.

Let A be $m \times n$, where the number of equations is m and the number of independent variables, n.

For $m < n$, even if all the equations are linearly independent, there are still fewer linearly independent equations than the number of independent variables. Therefore, the linear system has infinite solutions. For $m = n$, the linear system has a unique solution if and only if all the equations are linearly independent. For $m > n$, there must be linearly dependent rows. There can be at most n linearly independent rows and at least $m - n$ rows are linearly dependent. The linear system has a unique solution if and only if the number of the linearly independent rows, after removing those linearly dependent rows, matches n.

The number of linearly independent rows, in this context, is denoted by the rank of the matrix. (The formal definition of the rank of a matrix will be introduced in later part of the notebook.) For matrix A which is $m \times n$, $Ax = 0$ has unique solution if and only if $\text{rank}(A) = n$.

2.2 Vector and Matrix Calculations

Vector and matrix notations have been introduced in the earlier section. Basic vectors and matrices computation rules are introduced in this section.

2.2.1 Vector Calculations

Vectors are special matrix that has only one row or column, and they are known as **row vector** or **column vector** respectively. By saying "vector", we often refer to column vector. In the writing, a column vector can be denoted by the transpose of a row vector.

Equality. Two vectors are equal if and only if they are of the same size, and their corresponding entries are equal.

Vector addition. The sum of two vector exists only if they have the same size. The sum of two vector is obtained by adding corresponding entries. For example, consider $x = [x_1, \ldots, x_n]^T$, $y = [y_1, \ldots, y_n]^T$. The sum is given by

$$x + y = \begin{bmatrix} x_1 + y_1 \\ \vdots \\ x_n + y_n \end{bmatrix}$$

Subtraction works similarly.

Scalar multiplication. A vector multiplied by a scalar is obtained by multiplying each entry with the scalar. For example, consider $x = [x_1, \ldots, x_n]^T$ and c a scalar. The scalar multiple is given by

$$cx = \begin{bmatrix} cx_1 \\ \vdots \\ cx_n \end{bmatrix}$$

With sum and scalar multiple introduced, define the **weighted linear combination of vectors** as follows. Let $v_1, \ldots, v_p \in \mathbb{R}^{n\times 1}$ be p vectors. Let $c_1, \ldots, c_p$ be p scalars. Let $y \in \mathbb{R}^{n\times 1}$ be given by

$$y = c_1 v_1 + \ldots + c_p v_p$$

and y is known as a linear combination of $v_1, \ldots, v_p$ with corresponding weights $c_1, \ldots, c_p$. The value of y changes with different choice of weights. All probable values of y form the **span** of $v_1, \ldots, v_p$, often denoted by $\text{Span}\{v_1, \ldots, v_p\}$, which is obviously a subset of $\mathbb{R}^{n\times 1}$.

An interesting problem about span is to calculate the weights $c_1, \ldots, c_p$ given $v_1, \ldots, v_p$ and y.

Weighted linear combination of vectors is an important concept closely related to vector space (also known as linear space) which will be introduced formally in abstract algebra in later part of this notebook.

Dot product. The dot product of two vectors of the same size is the sum of multiplication of corresponding entries. For example, consider $x = [x_1, \ldots, x_n]^T$, $y = [y_1, \ldots, y_n]^T$. The dot sum is given by

$$x \cdot y = x_1 y_1 + \ldots + x_n y_n$$

Notice that the dot product of two vectors can be taken as the matrix multiplication of two matrix of size $1 \times n$ and $n \times 1$ in the form of $x^T y$ or $y^T x$. Matrix multiplication is introduced in the next section.

2.2.2 Matrix Calculations

A matrix can be written as a combination of rows or columns.

Matrix-vector product. Let $A \in \mathbb{R}^{m \times n}$ be a matrix, and let $x \in \mathbb{R}^{n \times 1}$ a vector. The multiplication Ax exists only if the number of columns in matrix A matches the number of rows in vector x, in this example, n. It is given by

$$Ax = \begin{bmatrix} a_1 & \ldots & a_n \end{bmatrix} \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} = x_1 a_1 + \ldots x_n a_n$$

where $a_i \in \mathbb{R}^{n \times 1}$ are the columns of A. Consequently, $Ax = b$, given A and b, has solution if and only if b is a linear combination of (linearly independent) columns of A.

Alternatively and equivalently, Ax can be interpreted as follows.

$$Ax = \begin{bmatrix} A_1 \\ A_2 \\ A_3 \end{bmatrix} x = \begin{bmatrix} A_1 x \\ A_2 x \\ A_3 x \end{bmatrix}$$

where A_i are the rows of A and $A_i x$ is the dot product.

3

Matrix Algebra

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Part II

Abstract algebra

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Abstract Algebra Basics

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Abstract algebra studies the principals and algorithms used in different algebraic systems. The basics of abstract algebra, including the scope and commonly seen definitions and concepts, are introduced in this chapter.

4.1 Scope of Abstract Algebra

Linear algebra has been introduced in the earlier part of this notebook. Linear algebra as well as other classic algebraic algorithms solves specific problems using algebra theories, whereas abstract algebra studies these theories. While classic algebra performs calculations on numbers, vector and matrices, **abstract algebra** studies the concepts, tools, derivations and logic we use in the calculation, and tries to explain why they work in the way they do. Ab-

stract algebra also develops new concepts, tools and algorithms that we can use to solve more complicated algebraic problems.

As an example, consider the following equation

$$Ax = y$$

where x, y are vectors and A a matrix. Solving x given particular A and y falls into the classic algebra domain. It is obvious that x does not necessarily exist or being unique for different y and A. Studying the general rules when x exists and when it is unique for a set of y and A becomes an abstract algebra problem.

As another example, consider the following expressions taught in elementary schools

$$\begin{aligned} a+b &= b+a \\ ab &= ba \end{aligned}$$

which are often used to demonstrate the commutative property of calculations (summation and multiplication, in this example). In classic algebra, they are considered as ground truth. In abstract algebra, however, the focus shifts to a more formal and generalized understanding of the property. We need to dig deeper into how commutative property is defined, and why it holds true for summation and multiplication, but not for some other operations such as division.

Here is another great motivating example. One of the most famous applications of abstract algebra is to study the analytical solution to the following polynomial equation

$$x^n + a_1 x^{n-1} + a_2 x^{n-2} + \ldots + a_n = 0 \tag{4.1}$$

where $n \geq 1$ is the order of the polynomial and $a_1, \ldots, a_n$ any arbitrary coefficients. The analytical solutions to (4.1) for $n = 1$ and $n = 2$ are obvious. With some effort, the analytical solutions for $n = 3$ and $n = 4$ were found in the 16th century. Since then, people have been struggling to find the analytical solution to the polynomial with fifth and higher orders.

In the 18th and 19th century, Euler, Lagrange and Gaussian attended this problem. Their conclusion was that there is no analytical solution to polynomial equations of fifth order or higher, but they could not give a very solid proof to the statement. Nevertheless, the methods they used inspired a lot of people that would work on this problem.

In the 19th century, Abel was able to prove that there is no analytical solution to general polynomial equations of fifth degree or higher with arbitrary coefficients (see Abel–Ruffini theorem). Furthermore, he discussed a set of special cases with specific coefficient patterns that can have analytical solutions, and he came up with a sufficient condition for a fifth order polynomial equation to have the analytical solution.

The necessary and sufficient condition for a fifth or higher order polynomial to have an analytical solution is finally fully discovered by the genius Galois at the remarkably young age of 18. Galois was able to create his theorem (known as the Galois theorem) and use it to find the ultimate answer to this problem that people have been studied for centuries. His theorem goes far beyond that. Galois theorem will find its usefulness in many areas to come, and eventually it becomes an important building block of abstract algebra.

In the remaining of this part of the notebook, abstract algebra including the contribution from Abel, Galois and other mathematicians are introduced.

4.2 Algebraic System

An **algebraic system** is a mathematical system consisting of a non-empty set and a series of operations defined on the set. Abstract algebra studies the properties of different algebraic systems. Depending on the properties of the algebraic systems, we can categorize them as "groups", "rings", "fields", "vector spaces", etc.

The commonly seen definitions and concepts used in algebraic system is introduced in the reminder of this section.

4.2.1 Domain and Mapping

Set is one of the most commonly used terms across different mathematical subjects. It is also one of the fundamental concepts in abstract algebra. A **set** usually refers to a collection of distinct objects. Given a set U and an object x, one and only one of the following two statements must be true:

- Object x is a member of set U, denoted by $x \in U$;
- Object x is not a member of set U, denoted by $x \notin U$.

However, notice that due to the Russell's paradox, it is challenging to give a rigorous mathematical definition to a set that fulfill the above statement. Therefore, we usually introduce set by its features, not its definition.

An algebraic system must contains a non-empty set. The set is known as the **domain** of the system.

Mapping, or **function**, is used to describe the association of elements in two sets. The two terms are used interchangeably in the this notebook.

An example of mapping is given as follows. Let A be a set, and $A_0 \subset A$ a subset of A. For any element $x \in A_0$, define mapping

$$i \quad : \quad A_0 \to A$$

where

$$i(x) \quad = \quad x$$

In this example, mapping i is called the **embedding mapping** from A_0 to A.

Let A, B be two sets, and $A_0 \subset A$ a subset of A. Let $f : A \to B$, and $g : A_0 \to B$. If $\forall x \in A_0$, $f(x) = g(x)$, function f is known as an **extension** of function g, and function g a **restriction** of function f (on A_0). This is denoted by $g = f|_{A_0}$.

Mappings can be chained together. For example, consider several mappings

$$\begin{aligned} f_1 &: A_1 \to A_2 \\ f_2 &: A_2 \to A_3 \\ f_3 &: A_3 \to A_4 \end{aligned}$$

With the above, we can denote $f = f_3 \circ f_2 \circ f_1$ a mapping from A_1 to A_4. Meantime, consider other mappings

$$\begin{aligned} g_1 &: A_1 \to B \\ g_2 &: B \to A_4 \end{aligned}$$

Clearly, $g = g_2 \circ g_1$ is also a mapping from A_1 to A_4. The mappings above can be illustrated intuitively using the **mapping diagram** in Fig. 4.1. Mapping

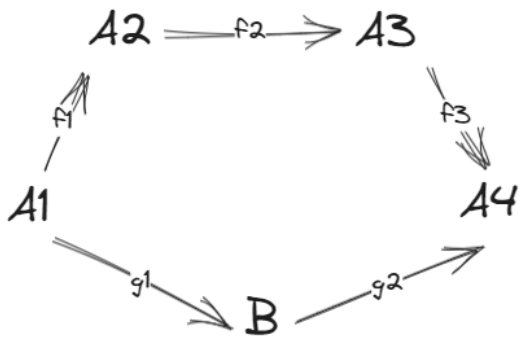


FIGURE 4.1: Mapping diagram that demonstrates f and g.

diagram can become handy with complicated mappings.

The embedding mapping, extension function and restriction function introduced earlier can also be represented by a mapping diagram as shown in Fig. 4.2, where $A_0 \subset A$ and $i : A_0 \to A$ a embedding mapping. Function g defined on the subset A_0 is a restriction of f defined on the superset, whereas f is an extension of g, i.e., $f(x) = g(x)$ for $x \in A_0$.

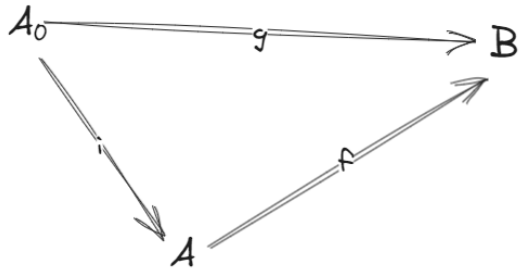


FIGURE 4.2: Mapping diagram of embedding mapping.

Multiple sets can be "combined" and "augmented" to form new sets. For example, the **Cartesian product** of two sets, A_1 and A_2, is defined as follows.

$$A_1 \times A_2 = \{(a, b) | a \in A_1, b \in A_2\}$$

where the tuple (a, b) can be interpreted as an ordered combination of elements in A_1 and A_2. The similar idea can be applied to Cartesian product of 3 or more sets.

4.2.2 Operation

Summation, multiplication, etc., are operations. In abstract algebra, we are more interested in the broad definition of operations from set perspective, instead of listing down individual operations and study how they work.

In an algebraic system, in addition to the domain, an operation must also be defined. An **operation** describes the rule of deriving or mapping from one or multiple elements to a new element in the domain.

Take binary operation as an example which maps two elements to one element. Let A, B, D be three non-empty sets. Let f be a mapping

$$f \quad : \quad A \times B \to D \tag{4.2}$$

Then f is called an **algebraic operation** from A, B to D. The operation can be denoted by $f(a, b)$ where $a \in A$ and $b \in B$. For convenience, binary operation is often denoted in the form of $a \oplus b$, $a \otimes b$, etc. Notice that there are conventions on how to use the symbols. For example, $+$, $-$, $\times$, etc., already have clear meanings.

The following is an example of operations. Let the domain of interest be $\mathbb{R}$, which is the real number set. Let v be a vector space (a formal definition of vector space will be given later; for now, take vector space as a set of vectors with summation operation defined on it) defined under $\mathbb{R}^n$. We can then define summation $+$ as a binary operation

$$+ \quad : \quad v \times v \to v$$

which indicates that the summation takes in two elements in the vector space, and generates a new vector that also belongs to the same vector space.

In the case where the domain and range of the operation come from the same set, i.e., in (4.2) $A = B = D$, the operation is said to be **closed** under this operation. In this example, the summation operation $+$ defined on v is closed, and we can simply say "$+$ is a binary operation defined on v".

Operations may have the following properties. Commonly seen ones are introduced below. Let binary operation $\oplus$ be defined on A, i.e., $\oplus : A \times A \to A$. If

$$a \oplus b \quad = \quad b \oplus a, \forall a, b \in A$$

then the operation is said to have **commutative property**. If

$$a \oplus (b \oplus c) \quad = \quad (a \oplus b) \oplus c, \forall a, b, c \in A \tag{4.3}$$

then the operation is said to have **associative property**, and (4.3) can be simply denoted by $a \oplus b \oplus c$. Let two binary operations defined on A, denoted by $\oplus$ and $\otimes$ respectively. If

$$a \otimes (b \oplus c) \quad = \quad (a \otimes b) \oplus (a \otimes c)$$

then the operation $\otimes$ is said to have the left-hand **distributive property** on operation $\oplus$. Similarly, right-hand distributive property can also be defined.

Commutative property, associative property and distributive property are commonly seen and widely discussed properties of operations. When operations have some of the three properties, simplified notation may apply. For example, if $\oplus$ operation has associative property, then

$$a^n \quad \equiv \quad \overbrace{a \oplus \cdots \oplus a}^{n}$$

can be used when there is no ambiguity. With this notation, it can be easily proved that for an operation that has both commutative and associative properties,

$$a^n b^n \quad = \quad (ab)^n$$

There are several different ways to represents the result of an operation. When the cardinality of the domain is finite, a simple way is to list all the input-output associations in a table. An example is given in Fig. 4.3 which exhaustively shows "AND" logical operation results. For operations defined on a domain with infinite cardinality, other methods must be used.

4.2.3 Relation

Consider the relation of two elements a and b. **Relation** is essentially a property defined on tuple (a, b). A straight forward way of defining a relation is to

AND	0	1
0	0	0
1	0	1

FIGURE 4.3: Result of "and" logical operation.

construct a set that contains some tuples, and if the specified tuple (a, b) is in the set, we say that a, b have the corresponding relation, and equivalently (a, b) has that associated property.

For example, consider $a, b \in \mathbb{Z}$ where $\mathbb{Z}$ denotes the integer set. Define relation

$$R \quad = \quad \{(a, b) \mid a - b = kn,\ k, n \in \mathbb{Z}, n > 1\} \tag{4.4}$$

If a, b are such that $(a, b) \in R$ in (4.4), we say a, b are congruent modulo n. This defines a **congruence relation** and it can be denoted by

$$a \quad \equiv \quad b \pmod{n}$$

Congruence relation is a commonly used relation in abstract algebra and number theory.

To summarize, let $a \in A$, $b \in B$. Let $R \subset A \times B$ be a subset of $A \times B$. the following statements can be considered equivalent.

- Two elements a and b have relation R;
- Tuple (a, b) has the property associated with relation R;
- Tuple $(a, b) \in R$;

and in that case we can use aRb to signify the relation. From above, we can see that each relation is associated with a subset R defined on the Cartesian product of the sets that the two elements belong to, and vise versa.

Relations may have the following properties. Commonly seen ones are introduced below. Let relation R be defined on $A \times A$. If $\forall a \in A$,

$$aRa$$

then R is said to be **reflexive**. If $\forall a, b \in A$,

$$aRb \quad \Rightarrow \quad bRa$$

then R is said to be **symmetric**. If $\forall a, b, c \in A$,

$$aRb, bRc \quad \Rightarrow \quad aRc$$

then R is said to be **transitive**. Finally, if $\forall a, b \in A$,

$$aRb, bRa \quad \Rightarrow \quad a = b$$

then R is said to be **antisymmetric**.

From the above definitions, we can see that the commonly seen "=" relation defined on $\mathbb{R} \times \mathbb{R}$ is reflexive, symmetric and transitive. Similarly, "$\leq$" is reflexive, transitive and antisymmetric.

4.2.4 Equivalence Relation and Equivalence Class

If a relation is simultaneously reflexive, symmetric and transitive, it is called an **equivalence relation**. The equal "=" relation and congruence relation introduced earlier are examples of equivalence relations.

We can define a **partition of a (non-empty) set** by grouping its elements into non-empty subsets in such a way that every element is included in exactly one subset, i.e.

$$\begin{aligned} A &= \bigcup_{i \in I} A_i \\ \text{s.t.} \quad & A_i \neq \varnothing \\ & \forall i, j \in I, i \neq j, A_i \bigcap A_j = \varnothing \end{aligned}$$

A partition of a set can form a new set $\{A_i\}$. Furthermore, we can define a relation $R \subset A \times A$ on top of that partition as follows

$$R = \{(a, b) \mid \exists i, a, b \in A_i\}$$

which intuitively means "the two elements are from the same subset in the partition" and it is clear that such R is an equivalence relation. A partition of a set A can determine an equivalence relation $R \subset A \times A$ using the above method. Vise versa, an equivalence relation $R \subset A \times A$ can also determine a partition of set A as follows. For $\forall a \in A$, define

$$[a] \quad = \quad \{b \in A | aRb\}$$

where R is the equivalence relation, and $[a]$ the **equivalence class** of a.

With the definition of an equivalence relation and a set of equivalence class, a partition of a set A can then be derived from the equivalence relation R by

$$A/R \quad = \quad \{[a] | a \in A\} \text{ (remove duplication)} \tag{4.5}$$

where A/R is the partition of set A (this can be proved easily) derived from equivalence relation R. The partition of set obtained using the above method is known as the **quotient set**.

There is a one-to-one correspondence between an equivalent relation and its quotient set. Given an equivalence relation, the following mappings can be defined.

$$\pi \quad : \quad A \to A/R,\ \pi(a) = [a]$$

which is known as the **canonical projection** or **quotient map**.

4.2.5 Congruent Modulo

Congruent modulo has already been introduced in (4.4). In that example, when two integers a and b are congruent modulo n, their equivalence relation persists even if $k_1 n$, $k_2 n$ are added or subtracted from both of them respectively.

The concept of congruent modulo can be further generalized in the context of operation and equivalence relation as follows. Let A be an non-empty set. Let $\oplus$ be a closed binary operation defined on A, i.e., $\oplus : A \times A \to A$. Let R be an equivalence relation defined on A, i.e., $R \subset A \times A$, and A/R its corresponding quotient set. Let $a_1, a_2, b_1, b_2 \in A$. If operation $\oplus$ and equivalence relation R satisfy the following condition

$$a_1 R a_2,\ b_1 R b_2 \quad \Rightarrow \quad (a_1 \oplus b_1) R (a_2 \oplus b_2) \tag{4.6}$$

then we say that R is **congruent modulo** $\oplus$. This is demonstrated by Fig. 4.4 the left plot. From Fig. 4.4 the left plot, congruent modulo can be intuitively interpreted as follows. If elements a_1, a_2 are congruent modulo $\oplus$, their equivalence relation persists even if operation $\oplus b_1$, $\oplus b_2$ are applied to them respectively, so long as b_1, b_2 are also congruent modulo $\oplus$.

If R is congruent modulo $\oplus$ indeed, we can define an operation on the quotient set A/R as follows. Let $[\cdot]$ denote the equivalence class of an element in A. Define $\bar{\oplus}$ as an operation on A/R as follows.

$$[a] \bar{\oplus} [b] \quad = \quad [a \oplus b],\ a \in [a],\ b \in [b] \tag{4.7}$$

where a, b are arbitrarily selected elements in the equivalence class $[a]$ and $[b]$ respectively, as shown in Fig. 4.4 right plot. Notice that $[a \oplus b]$ should be well-defined regardless of the choice of a and b so long as they are selected in their associated equivalence class $[a]$ and $[b]$ respectively since R is congruent modulo $\oplus$. In this context, $\bar{\oplus}$ is known as the **included operation** or **quotient operation** of $\oplus$ on the quotient set A/R.

To conclude, in the algebraic system with non-empty set A and an operation $\oplus$ (denoted by $\{A; \oplus\}$), if there is a equivalent relation R which is congruent modulo $\oplus$, we can construct a new algebraic system $\{A/R; \bar{\oplus}\}$, where A/R is the quotient set of A on R from (4.5), and $\bar{\oplus}$ the included operation of $\oplus$ on A/R from (4.7).

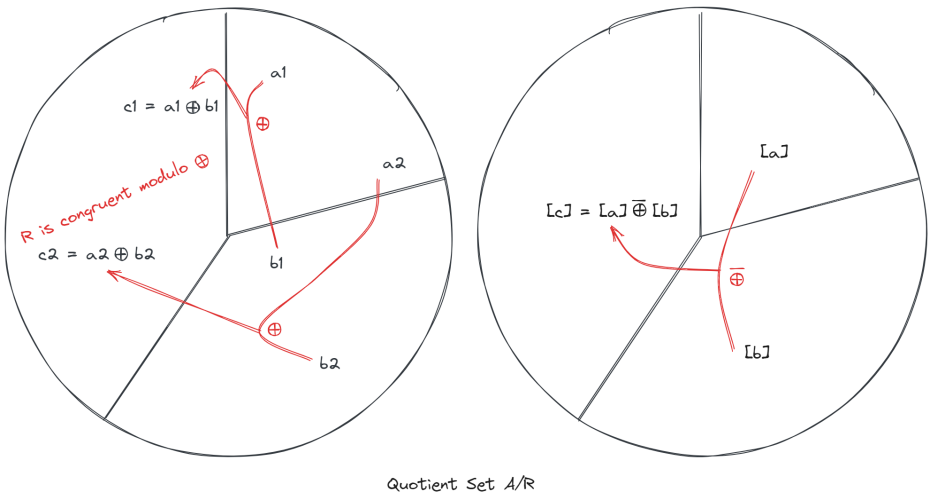


FIGURE 4.4: A relation is congruent modulo an operation.

4.3 Semi-Group and Group

As introduced earlier, a general algebraic system shall contain a set and one or more operations defined on the set. The operations can be unary (involving one operand), binary (involving two operands), ternary (involving three operands), etc. Relations are defined as sets of tuples, with each element of the tuple coming from the set of interest. Likewise, relation can also be binary, ternary, etc.

Special properties can apply to both operations and relations. Binary operations, for instance, can exhibit commutativity, associativity, and distributivity. Binary relations, on the other hand, can be reflexive, symmetric, and transitive. If a relation is simultaneously reflexive, symmetric and transitive, it is called an equivalent relation. Each equivalent relation is corresponded with a quotient set which is a partition of the original set by equivalence classes of the relation. When a relation is congruent modulo an operation (recall (4.6) and Fig. 4.4), we can establish a new algebraic system. This system comprises the quotient set and a well-defined induced operation on that set. The new system retains structural properties from the original set, reflecting the consistency and compatibility of the congruence relation with the operation.

Moving forward, our focus will shift to specialized algebraic systems characterized by distinctive properties. This exploration will pave the way to understanding the fundamental algebraic structures such as semi-groups, groups, and rings. We will see the critical impact the operation possess over the alge-

braic system. Even with the same set, different operations often leads to very different features of the algebraic system.

4.3.1 Semi-Group

Let S be a non-empty set, and $\oplus : S \times S \to S$ a closed binary operation defined on S. If $\oplus$ has associative property, i.e. $a \oplus (b \oplus c) = (a \oplus b) \oplus c$, then algebraic system $\{S; \oplus\}$ (for simplicity, S, when there is no ambiguity) is called a **semi-group**.

Examples of semi-group include $\{\mathbb{N}; +\}$, $\{\mathbb{N}; \times\}$ where $\mathbb{N}$ refers to the (positive) natural numbers $1, 2, 3, \ldots$

Another example is given as follows. Let A be an non-empty set, $\mathcal{M}(A)$ the set of all the closed mappings defined on A, and $\circ$ the compound operation. Then $\{\mathcal{M}(A); \circ\}$ is also a semi-group. Additionally, $\{\mathcal{P}(A); \bigcup\}$, $\{\mathcal{P}(A); \bigcap\}$ are also semi-groups, where $\mathcal{P}(A)$ represents the power set of A.

Is Zero Included in Natural Numbers?

The definition of natural numbers may or may not include zero "0" depending on the context. In the context of number theory and abstract algebra, natural numbers often exclude zero.

4.3.2 Monoid

Consider semi-group $\{S; \oplus\}$. If $\exists e \in S$ so that

$$e \oplus a \quad = \quad a, \ \forall a \in S$$

then e is known as the **left identity**. Similarly, we can define **right identity**. Notice that a semi-group may have many distinct left and right identities.

If an element e is both left identity and right identity, it is called the **identity element**. If a semi-group has an identity element, the identity element must be unique. This can be easily illustrated using proof by contradiction as follows. Assume that e_1, e_2 are two distinct identity elements, thus

$$e_1 \oplus e_2 = e_1 \quad \text{and} \quad e_1 \oplus e_2 = e_2$$

indicating $e_1 = e_2$, which contradicts with the assumption.

A semi-group with the identity element is knows as a **monoid**. From the definition, we know that $\{\mathbb{N}; \times\}$, $\{\mathcal{M}(A); \circ\}$, $\{\mathcal{P}(A); \bigcup\}$ and $\{\mathcal{P}(A); \bigcap\}$ are all monoids, with their identity elements being 1, $f : A \to A, f(a) = a$, $\varnothing$ and A, respectively.

4.3.3 Group

Let $\{S; \oplus\}$ be a monoid with identity element e. If for $a \in S$, there is

$$a' \oplus a \quad = \quad e$$

then a' is called the left inverse of a. Likewise, right inverse can be defined. Notice that an element may have many left and right inverses.

If a' is both the left and the right inverse of a, then a' is known as the inverse of a, ands a' must be unique. The proof can be obtained similarly using proof by contradiction. The inverse of a is often denoted by a^{-1}.

If in a monoid $\{S; \oplus\}$, there is the inverse for all its elements, the monoid is called a group, usually denoted by $\{G; \oplus\}$, or simply G when without ambiguity.

The relationship of general algebraic system, semi-group, monoid and group are demonstrated in Fig. 4.5.

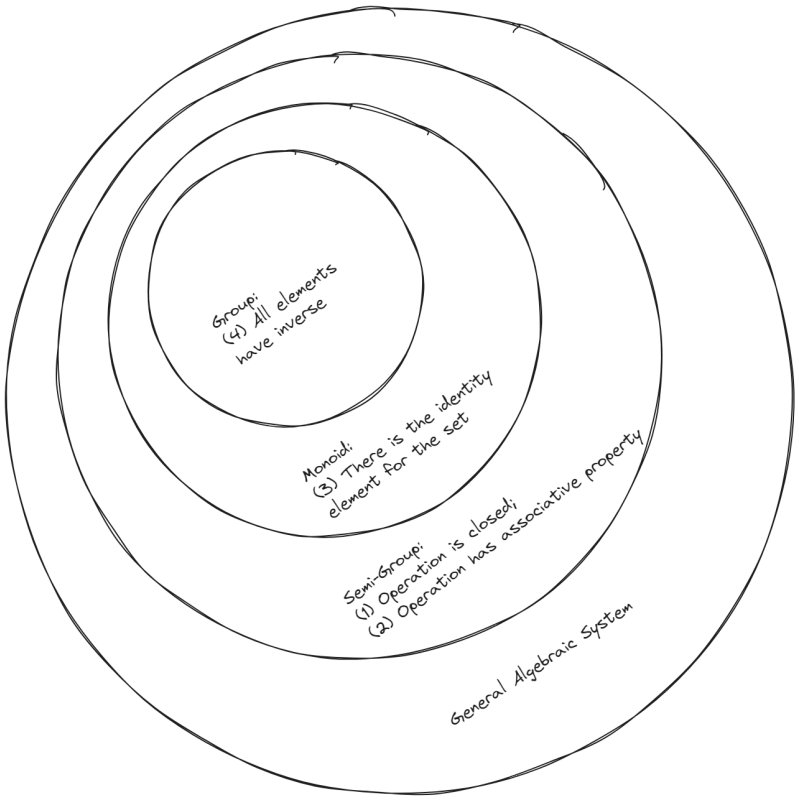


FIGURE 4.5: General Algebraic System, Semi-Group, Monoid and Group.

Notice that group does not pre-assume commutativity. If a group's operation is commutative, it is called an **abelian group** named after Niels Henrik Abel. Examples of abelian group include $\{\mathbb{Z}; +\}$, $\{\mathbb{R}; +\}$, $\{\mathbb{C}; +\}$ where $\mathbb{Z}$, $\mathbb{R}$ and $\mathbb{C}$ represent integer set, real number set and complex number set, respectively. Additionally, $\{\mathbb{R}^*; \times\}$, $\{\mathbb{C}^*; \times\}$ are also abelian groups, where $\mathbb{R}^*$, $\mathbb{C}^*$ denotes the corresponding sets excluding zero.

We know that $\{\mathcal{M}(A); \circ\}$ introduced earlier is not a group. This is because not all mappings defined on a set A necessarily have an inverse. However, if define $\{\mathcal{S}(A); \circ\}$ where $\mathcal{S}(A)$ represents the set of all bijections (invertible mappings) on A, $\{\mathcal{S}(A); \circ\}$ becomes a group. Notice that $\mathcal{S}(A)$ is not necessarily an abelian group.

To conclude, a group must meet the following requirement:

- There is a non-empty set G;
- There is a closed binary operation $\oplus : G \times G \to G$;
- The operation $\oplus$ has associative property, i.e. $\forall a, b, c \in G, (a \oplus b) \oplus c = a \oplus (b \oplus c)$;
- There is the identity element, i.e. $\exists e \in G$, $\forall a \in G, e \oplus a = a \oplus e = a$;
- There is the inverse for all elements, i.e., $\forall a \in G$, $\exists a^{-1}$, $a \oplus a^{-1} = a^{-1} \oplus a = e$.

Furthermore, if

- Operation $\oplus$ has commutative property, i.e. $\forall a, b \in G, a \oplus b = b \oplus a$;

the group is called an abelian group.

It can be proved that if a semi-group has left identity and all its elements have left inverse, the left identity must also be its right identity and all its elements must also have right reverse, thus making the semi-group a group. The proof is given below. Let e be the left identity of semi-group $\{S; \oplus\}$, while b be the left inverse of any arbitrary element $a \in S$. Let c be the left inverse of b.

$$\begin{aligned} a \oplus b &= e \oplus a \oplus b \\ &= c \oplus (b \oplus a) \oplus b \text{ (associativity)} \\ &= c \oplus (e \oplus b) \text{ (associativity)} \\ &= c \oplus b \\ &= e \end{aligned}$$

Therefore, b is also the right inverse of a. Furthermore,

$$\begin{aligned} a \oplus e &= (a \oplus b) \oplus a \text{ (associativity)} \\ &= e \oplus a \\ &= a \end{aligned}$$

Therefore, e is also the right identity. All the above derivations also apply when a semi-group has right identity and all its elements have right inverse.

4.3.4 Properties of Semi-group and Group

Commonly used properties and notations of semi-group and group are introduced in this section.

If $\{S;\oplus\}$ is a semi-group and $a,b \in S$, then the following criteria can be used to further determine that $\{S;\oplus\}$ is a group.

- For $\forall a,b \in S$, if both $a \oplus x = b$ and $x \oplus a = b$ have solution, then S is a group.
- If $|S| < \infty$ (cardinal number of S is finite), and $\forall a,b,c \in S$, $a \oplus c = b \oplus c \Rightarrow a = b$, $c \oplus a = c \oplus b \Rightarrow a = b$, then S is a group.

If G is a group and $a,b,c \in G$, then the following properties apply.

- $a \oplus c = b \oplus c \Rightarrow a = b$;
- $c \oplus a = c \oplus b \Rightarrow a = b$;
- $a \oplus x = b$ has the unique solution $x = a^{-1} \oplus b$; $x \oplus a = b$ has the unique solution $x = b \oplus a^{-1}$;

Given group $\{G;\oplus\}$ and $a \in G$. The following notations apply.

$$\begin{aligned} a^n &\equiv \overbrace{a \oplus \cdots \oplus a}^{n} && (4.8)\\ a^0 &\equiv e && (4.9)\\ a^{-n} &\equiv \overbrace{a^{-1} \oplus \cdots \oplus a^{-1}}^{n} && (4.10) \end{aligned}$$

where n is a positive integer, e the identity element of G and a^{-1} the inverse of a. With this notation, instead of (4.8), (4.9) and (4.10), the following properties apply.

- $a^m a^n = a^{m+n}$
- $(a^m)^n = a^{mn}$

When studying abelian groups, "+" is most commonly used to represent the operation, and the following notations apply.

$$\begin{aligned} na &\equiv \overbrace{a + \cdots + a}^{n}\\ 0a &\equiv e && (4.11)\\ (-n)a &\equiv \overbrace{(-a) + \cdots + (-a)}^{n} \end{aligned}$$

where $-a$ is used to denote the inverse of a. In this case, the identity element e in (4.11) is often denoted by "0". Note that in this context, 0 represents the

identity element of the abelian group and does not have to be the numerical number zero. When the group G is $\mathbb{Z}$, $\mathbb{R}$ or $\mathbb{C}$, and the operation is indeed summation, the identity element is indeed numerical zero.

The idea of group goes beyond number theory and mathematics. Generally speaking, any system that can move from states to states with movements revertible can be described by a group or something similar. When a system shows symmetry, it can probably be described by a group.

4.3.5 Order of Elements in a Group

Let G be a group, e its identity element, and $a \in G$. If there exists a smallest positive integer n so that $a^n = e$, we say that the **order** of a is n. If no such n exists, the order of a is infinity. The identity element e is the only element that has the order of 1. For an element with order n, its inverse also has the order of n.

Let G be a group, e its identity element, and $a \in G$. Consider the sequence of a^n as follows.

$$\ldots \quad a^{-3} \quad a^{-2} \quad a^{-1} \quad a^0(e) \quad a^1 \quad a^2 \quad a^3 \quad \ldots$$

If a has the order of infinity, then $a^m \neq a^n$ for any integers $m \neq n$, and vise versa. This can be easily proved with proof by contradiction. If a has the order of d, then $a^m = a^n$ if and only if $m - n = kd, k \in \mathbb{Z}$.

If $a \in G$ has order d, then $a^k, k \neq 0$ has order $d/(d, |k|)$, where $(d, |k|)$ is the greatest common divisor of d and $|k|$. This implies that a^k has order no more than d, and it has order d if and only if $(d, |k|) = 1$.

If $a, b \in G$ has order m and n respectively, and $ab = ba$, then ab and ba have the order of $[m, n]$, where $[m, n]$ stands for the least common multiple of m and n.

4.4 Subgroup and Quotient Set

Subgroup and quotient set are both fundamental concepts in group theory. They not only form an important part of abstract algebra by themselves, but also help with better understanding group and its properties. Notice that quotient set has been mentioned earlier during the introduction of the equivalence relation and equivalence class. More details are given here.

4.4.1 Subgroup

Let $\{G; \oplus\}$ be a group, and $H \subseteq G, H \neq \varnothing$ is a non-empty subset of the group. If $\{H; \oplus\}$ forms a group under the same operation $\oplus$, then H is called

a **subgroup** of G, denoted by $H \leq G$. Notice that if $H \neq G$, H is called a **proper subgroup** of G and can be denoted by $H < G$. That implies that

- The operation $\oplus$ which is closed in G is also closed in H;
- The identity element of G, denoted by e, is in H;
- For every element $a \in H$, its inverse a^{-1} is also in H.

Here is an example of a subgroup. Consider group $\{\mathbb{R}^*; \cdot\}$ where $\mathbb{R}^*$ denotes real number set excluding zero, and $\cdot$ the multiplication operation. Then group $\{\mathbb{R}^+; \cdot\}$ where $\mathbb{R}^+$ denoting positive real number set is a subgroup of $\{\mathbb{R}^*; \cdot\}$.

Here is another example of a subgroup. The **general linear group** of degree n consists of $n \times n$ invertible matrices with the operation of ordinary matrix multiplication. It is denoted by $GL_n(F)$ where F is the field / ring of interest such as the complex numbers field, integer ring, etc. More about field and ring are introduced in later sections and chapters. Obviously, $GL_n(F)$ is a group. The **special linear group** of degree n, denoted by $SL_n(F)$ which consists of matrices with determinant of 1, therefore, is a subgroup of $GL_n(F)$.

In practice, an invertible matrix A is equivalent with a linear transformation $x \mapsto Ax$. Therefore, $GL_n(F)$ can be taken as a group of linear transformation as well (hence the name), and from that perspective it is a subgroup of the **general affine group** $x \mapsto Ax + b$.

The following criteria can be used to determine whether a subset is a subgroup under the operation. Let $\{G; \oplus\}$ be a group, and $H \subseteq G$ a non-empty subset. The following three statements are equivalent.

(i) $H \leq G$;

(ii) $\forall a, b \in H$, $a \oplus b \in H$, $a^{-1} \in H$;

(iii) $\forall a, b \in H$, $a \oplus b^{-1} \in H$.

Furthermore, if $|H| < \infty$, the following two statements are equivalent.

(i) $H \leq G$;

(ii) $\forall a, b \in H$, $a \oplus b \in H$.

The proof is neglected in this notebook.

Subgroup has the following properties.

- If $H_1 < G$, $H_2 < G$, $H_1 \bigcap H_2 \neq \varnothing$, then $H_1 \bigcap H_2 < G$;
- If $H_1 < G$, $H_2 < G$, then $H_1 \bigcup H_2$ is not necessarily a subgroup of G.

4.4.2 Coset of a Subgroup

Let $\{G;\oplus\}$ be a group, $H < G$ a subgroup, and $a \in G$. Define **left coset** of H about a, aH, as follows.

$$aH \quad = \quad \{a \oplus h | h \in H\}$$

Similarly, the **right coset** Ha can be defined.

Let $\{G;\oplus\}$ be a group and $H < G$. Define the following relation for any two elements in G as follows.

$$aRb \quad = \quad \left\{(a,b)\middle| a^{-1} \oplus b \in H\right\} \tag{4.12}$$

It can be proved that the above relation is an equivalence relation and the equivalence class of a, $[a]$, is nothing but $[a] = aH$. The proof is neglected here.

This theorem associate a subgroup H with an equivalence relation (4.12).

4.4.3 Quotient Set of a Subgroup

Recall that in an algebraic system an equivalence relation is corresponding with a quotient set. Given a subgroup $H < G$, an equivalence relation (4.12) can be defined, hence the quotient set G/R. This quotient set is known as the left quotient set of subgroup H.

Part III

Number Theory

Bibliography

[1] Nathan Jacobson. *Basic algebra I and II.* Courier Corporation, 2012.